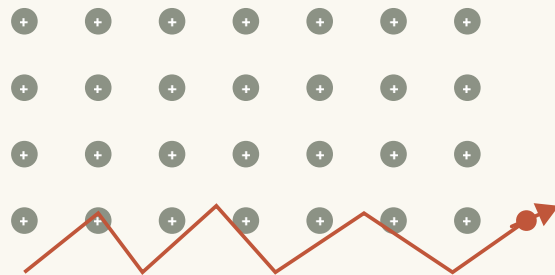


Resistance of a Conductor

Same metal, same electricity — yet thickness, length and material all matter.

OPENING QUESTION

Why are household wires and bare high-voltage lines so different — and exactly how much does each factor matter?



drifting electron scatters off ions

CONTEXT

Sometimes resistance is the whole point.

Every conductor resists current; resistors put that resistance to deliberate use.

01 · Every conductor resists

Wires, filaments, even your body oppose current to some degree.

02 · Resistors are deliberate

Components that provide a precise, reliable resistance.

03 · Why we need them

They set currents and divide voltages — the workhorses of every circuit.

MICROSCOPIC PICTURE

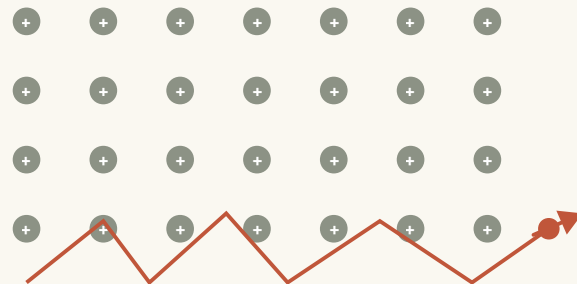
Where does resistance come from?

Inside a metal, free electrons drift through a lattice of vibrating positive ions.

Each collision with an ion disturbs the drift — the accumulated effect is what we call **resistance**.

WATER ANALOGY

An obstructed pipe needs more pressure for the same flow.



drifting electron scatters off ions

DEFINITION

Putting a number on it.

Resistance compares the voltage across a conductor with the current through it.

$$R = \frac{U}{I}$$

01 · A ratio at one point

U and I measured together, at the same moment. Unit: ohm, $1 \Omega = 1 \text{ V/A}$.

02 · A property of the conductor

Set by material, dimensions and temperature — not by the circuit.

03 · Universal

$R = U/I$ defines resistance for every element, linear or not.

LAW

Ohm's law — and its limits.

Current is proportional to voltage — for some conductors, at steady temperature.

WHERE IT HOLDS

Metallic conductors and electrolyte solutions: $I = U/R$ with R constant.

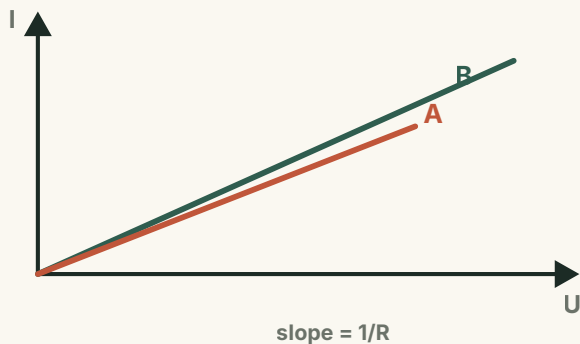
WHERE IT FAILS

Gas-discharge tubes, semiconductor devices, motors and electrolytic cells.

KEEP APART $R = U/I$ **defines** resistance always; Ohm's law is a special property of some materials.

Reading resistance from a graph.

Check the axes before you compare slopes — the same conductors look opposite.



U-I GRAPH

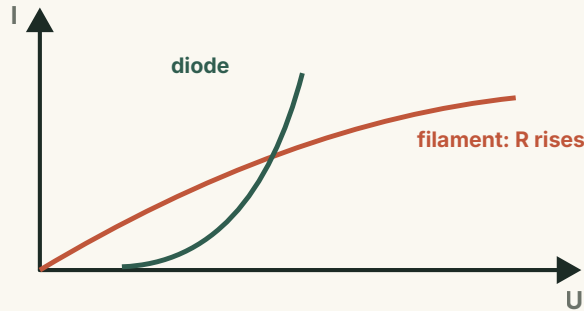
Slope = R . The steeper line has the larger resistance.

I-U GRAPH

Slope = $1/R$. The steeper line has the smaller resistance.

Non-ohmic elements.

$R = U/I$ still gives the resistance at every point — what is lost is only the proportionality.



FILAMENT LAMP

Current heats the filament; R rises and the curve bends toward the U axis.

SEMICONDUCTOR DIODE

Resistance collapses as U rises; the curve bends toward the I axis.

QUICK CHECK

Quick check: reading R off a curve.

The graph shows I against U for a conductor. Which statements are correct?

A $R = U/I$ cannot be used for a non-linear element

B ✓ At $U = 5\text{ V}$ its resistance is about $5\ \Omega$

C Its resistance increases as the voltage rises

D ✓ Its resistance decreases as the voltage rises

ANSWER

B · D — read $R = U/I$ at each point: $5\text{ V} / 1.0\text{ A} = 5\ \Omega$, but $10\text{ V} / 2.5\text{ A} = 4\ \Omega$. The ratio falls, so R decreases.

EXPERIMENT

What determines the resistance of a wire?

Four suspects — so we change one at a time and hold the rest fixed: the control-variables method.

01 · Length

How far the electrons must travel.

02 · Cross-section

How wide the channel is.

03 · Material

Which metal, and its structure.

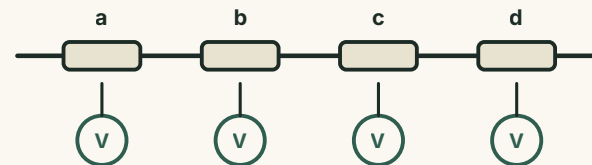
04 · Temperature

How vigorously the lattice vibrates.

One circuit, three fair comparisons.

Conductors a–d in series, a voltmeter across each — current is the same everywhere.

- 1 a vs b**
Only the length differs.
- 2 a vs c**
Only the cross-section differs.
- 3 a vs d**
Only the material differs.



conductors a–d in series, a voltmeter across each

The resistance law.

Resistivity ρ is set by the material; L is length and S is cross-sectional area.

RESISTANCE OF A UNIFORM WIRE

$$R = \rho \frac{L}{S}$$

Double the length $\rightarrow$ double R . Double the cross-section $\rightarrow$ halve R .

Ag $1.6 \times 10^{-8} \Omega \cdot \text{m}$

Cu $1.7 \times 10^{-8} \Omega \cdot \text{m}$

Al $2.9 \times 10^{-8} \Omega \cdot \text{m}$

W $5.3 \times 10^{-8} \Omega \cdot \text{m}$

Nichrome $\approx 110 \times 10^{-8} \Omega \cdot \text{m}$

GOING FURTHER

Resistivity moves with temperature.

The same material property behaves differently across classes of solids.

01 · Metals

ρ rises steadily with T — the basis of resistance thermometers.

02 · Semiconductors

ρ falls sharply as T rises — thermistors exploit this.

03 · Superconductors

Below a critical temperature, resistivity vanishes entirely.

STANDARD RESISTORS Alloys such as manganin barely move with temperature — exactly what you want.

WORKED EXAMPLE 1

Worked example 1: resistance from a curve.

Lines a and b are I–U characteristics; dashed c is the tangent to b at $U = 1\text{ V}$, parallel to a.

A At $U = 1\text{ V}$, the resistance of b is $5\ \Omega$

B ✓ At $U = 1\text{ V}$, a and b have equal resistance

C ✓ The resistance of b rises as U rises

D At $U = 3\text{ V}$, a and b have equal resistance

ANSWER B · C — resistance is U/I at the point, not the tangent slope; b's ratio grows as U rises.

WORKED EXAMPLE 2

Worked example 2: testing the factors fairly.

In the series circuit, b differs from a only in length, c only in cross-section, d only in material.

A ✓ Without an ammeter, no quantitative study is possible

B a vs b: the longer conductor gives the larger reading

C a vs c: the thicker conductor gives the larger reading

D a vs d: a larger reading means the better conductor

ANSWER A — voltmeter readings compare resistances, but a current measurement is needed for numbers. (The longer and thinner conductors read larger; a larger reading means a poorer conductor.)

WORKED EXAMPLE 3

Worked example 3: thinking straight about resistivity.

Which statement about the resistivity of materials is correct?

A Cut a wire in three — each piece has a third of the resistivity

B Every material's resistivity rises with temperature

C ✓ An alloy's resistivity exceeds the pure metals composing it

D Larger resistivity means more opposition to current

ANSWER

C — ρ is a material property, independent of size. It does not always rise with T (semiconductors fall), and opposition depends on geometry too.

WORKED EXAMPLE 4

Worked example 4: one block, two resistances.

A copper block has edges $a > b > c$. Send current lengthwise (edge a) or through the thickness (edge c).

A 1 : 1

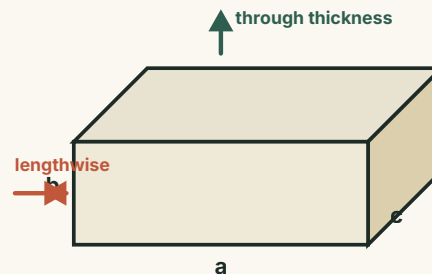
B ✓ $a^2 : c^2$

C $a^2 : b^2$

D $b^2 : c^2$

ANSWER

B — $R = \rho L/S$. Lengthwise $L = a$, $S = bc$;
through-thickness $L = c$, $S = ab$. The ratio
is $a^2 : c^2$.



● SUMMARY

What to take away.

01 · Definition

$R = U/I$ — any conductor, any operating point. A ratio, never a tangent slope.

02 · Ohm's law

$I = U/R$ with R constant — a straight I - U line. Metals and electrolytes only.

03 · Resistance law

$R = \rho L/S$ — longer means more, thicker means less, material sets the scale.

04 · Resistivity

A material property that moves with temperature — up for metals, down for thermistors.